MUSHROOM PRODUCTION

By

EVANGELIST

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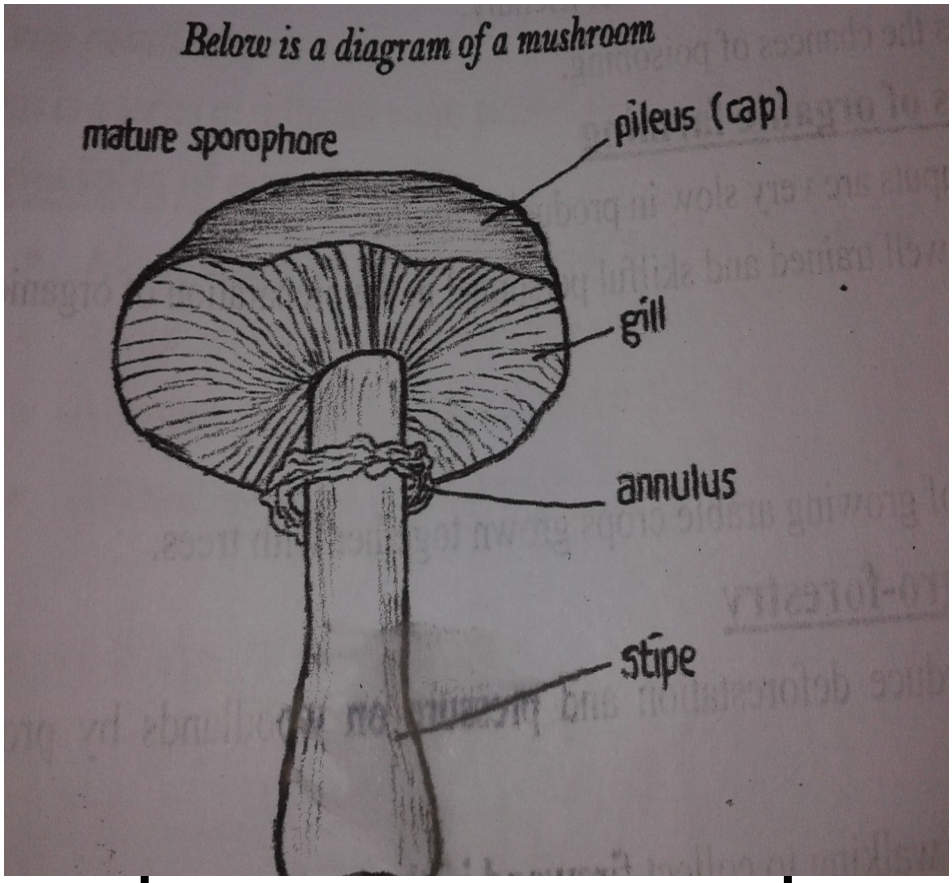
Mushroom is a freshy spore bearing fruiting body of a fungus produced above the ground or on its food source .

PARTS OF A MUSHROOM



1- Stipe

This provide support to the cup (Stem ya bowa imatchedwa Stipe)

2- Cap

This suport and protect gills and pores where spores are produced (And Cap imateteza ma gills mdipo nthangala zimapangidwa mmenemo)

3- Gills

This produce spores called basidoispores

4- Spores

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These are produced under the gills of the mushroom

5- Ring

This grows between the edge of the cup and stipe

IMPORTANCE OF MUSHROOM

- source of proteins vitamins and minerals which are important in the human body.
- source of income to farmers
- source of employment
- source of foreign exchange after being exported outside the country

- assists in crop diversification and provide a better way of using resources such as animal wastes

(Ok apa am explaining kufunikila kwa mushroom)

-Timapezamo ma protein vitamin zomwe nzofunikila mmatupi mwathu

-Bowa amapezesa ndalama kwa alimi akagulisa

-Bowa amabwelesa ma dollars tikagulisa kunja

-Bowa amafuna malo ochepa kulimidwa choncho nhakhale anthu taswana kwambiri ulimiwu ngosavuta kuupanga

-Bowa amathandizaso kuonjezela ulimi pamwamba pa zomwe timalima nkale komaso umagwilisa ntchito ngakhale ndowe

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Mitundu ya bowa omwe amalimidwa mmalawi

1- Oyster mushroom

2- Button mushroom

OYSTER MUSHROOM

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It requires low

technical farming practices as such it is easy to be grown by most malawian farmers.

Management practices for oyster mushroom production

(A) Site selection*

- requires temperature ranging from 12degrees celcius to 24 degrees

(b)

construction of incubation and production house* .

- good house constructed from a wooden frame work and covered with a plastic sheet

- thatched by grass to maintain room

(C) the house should have light

- production house should have an incubation room momwe light simafumidwa.

-If an incubation room isnt included in the production house , a black plastic sheet is used to cover the bags

-Nyumba yomwe mmazalamo bowa mmayenela mukhale light koma mmayenelaso mukhale kachipinda kamdima momwe mmafesedwa mbeu

(d) subrate preparation

- subrate is a material in which mycelium colony growz or attached . It is where organic materials are connected into a medium for growth of a mushroom

Materials used in the preparation of substrate

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- Finely choped dry maize stalks , rice straw or banana leaves with suprementation.

- shrewd maize cobs with or without supprementation

- cotton waste with or withought supprementation

- saw dust with supprementation

What is substrate ?

- A material in which mycelium colony grow

-Ok pofuna kupanga substrate timagwirisa ntchito zinthu izi

- mapesi achimanga odula dula kapena zanfhochi or madeya ampunga

NB: supprements (zoonjezela) commonly used are madeya(15-20%)

Calcium calbonate(1-2%)

Water hyacinth (10-20%)

-Akati sawdust with supprementation ndekuti utuchiwo paokha sumagwira unless aonjezele chinthu china .

-So akati without supprement ndekuti material amenewo paokha ngokwanila ali ndi nthaka

PROCEDURE

(1) choped materials are dipped into powdered soap solution. It is left for five to ten minutes in the container .

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-The substrate is rinsed into clean water to remove soap bubbles

-Mapesi akadulidwa amaviikidwa mmazi othila surf kwa five to ten minutes kenako zimavuulidwa nkuziviika mmazi oyela kuchosa thovu

(2)- afta this substrate is put in clean sack over night to drain.

(3)- moisture then checked by sqeeze test before packing in plastic growing bags.

(4)- well prepared substrate are supremented with suprements . They help to improve yield of mushroom

(Mwaona apa akuti substrate yabwino ndiyomwe aikako suprement chifukwa imaonjezela nthaka nkuchulusa bowa omela)

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-Moonjezela pa surf paja ntchito yake nkufewesa materials aja achina mapesi komaso surf ndi disinfectant amapha tima bacteria

(5)- suprement ya Lime imathandiza kupanga improve soil PH komaso soil aeration

(6)- It is either pasteurised or not.

What is pasteurisation?

-It means exposing something to higher temperature to destroy micro organizms

(6)- saw dust from hardwood or broadleaved trees are used . It is wetted and incubated overnight with suprements added and then pasteurised

(7)- suitable substrates for oysters mushroom should be prepared using shredded maize stalks and husks . This is the cheapest way of producing planting spawns

What are spawns ?

-Mushroom spawn is simply any substance that has been inoculated with mycelium, the vegetative growth of a fungus.

*Sources of Spawn

They are produced in Bunda college, NRC, Vumbwe research station, Biology departmemt at chaco

Subsrate treatment

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- afta preparatiin of substrate theybare packed in home made sterilizer made of the drumwith holes .

-Afted loading it is then tightly coverd with plastic sheet or sealable cover with an opening

- pasteurise for 1 hr and put the substrate on the plastic sheet to cool down up to 40degrees

- substrate are then packed into bags

Mushroom seeding

This is known as spawning and it involves addition of mushroom seeds into the bags containing subsrate.

- once the seedz are put into bags the mouth is tied . It is done to enclosed place where there is no airflow.

-spawned bags are coverd by black plastic sheet until they are fully colonised

- it takes 14-40 days at 24degrees for full colonisation.

Spawning means adding mushroom seeds into the bags containing substrate

-Tinganene ngati kuzala nthangala za bowa mu substrate

Mushroom fruiting management

- its done within production house

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- spawned substrate are transfered into fruiting house wjere enaf light is provided for the mushroom to start formin.

- 1kg of choped maize produces 1.8kg to 1.9kg of oyster mushroom

- temperature of 18-25 and humidity of 80-90 % shud be maintained

-Relative humidity is maintained by applying water several times a day on the floor and walls

- insufficient fresh air results into accumulation of CO_2 and this affects fruiting process

Pest and disease control in mushroom production .

(1) Green moulds

-Produces sticky spores and carried by flies mice n pickers hands to infected areas

Symptoms

- green moulds on dead mushroom, stump and on substrate

control

- disinfect the spawning house
- dispose away all the infected substance
- using deosan super or adecol for disinfection
- Matenda oyamba abowa ndi green mould ndipo bowa akagwidwa amaonesa zinthu za green pa bowa ndi pa substrate

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-Kupewa kwake nkuchosa zomwe zagwidwa ndi matenda nkukataya kutali

Komaso kupopelamo mankhwala omwe talemba apowo

(2)- cobweb moulds

-caused by too little air movement high humidity and high temp in the house

Control

- remove stumps and dead mushroom regulary

- spraying 0.5% formaldehyde solution on the spots.

(3)Penicillium spp

Caused by improper substrate pasteurisation or unsterile condition during spawning of sterilized substrates.

-produces spores which look like smoke

-reduces yield up to 80%

Control.

-proper hygien during spawning

-removing all infected substrates immediately.

Pests of Oyster Mushroom.

These include snails, phorid , and scarid flies , beetles and mites..

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Harvesting Oyster Mushroom

This is done by plucking mushroom when it is fully grown up

It takes about 5-9days for the flush to come out after opening the bag.

(2) BUTTON MUSHROOM

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There are two varities of button mushroom

TNS 1

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- Mbeu iyi imafuna temperature ya 16 to 22 degrees celcius.

-it is white in colour

-umatha kukolola zochuluka 15kgs per square KM

TNS2

Variety iyi imafuma temperature ya 18 to 22 degrees celcius .

- its bigger in size than TNS1

- white in colour with potential yield of 15kg per square KM

- temperatures are within march and july and this isthe right time for growing TNS2.

Housing Button mushroom

pomanga nyumba ya bowa iyu izi zuenela kusatidwa.

-Humidity

-Temperature

-Ventilation

-Moisture

Qualities of a good house for button mushroom*

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(1)Siting

- house should be free from obstructions

- away from ventilation

(2)Frame work

Should be made of wooden framework covered by plastic sheets to contrl humidity

-Should be thatched by grass to control temperature

-zida zogwilisila ntcito nazi; wooden poles, bamboos, timber nails, thatching grass and wire or plastic gauze.

(3) Dimenstion of the house

-Yaing'ono izikhala 5m × 3m × 2m.

-yaikulu izikhala 7m× 5m× 3m

Inside the house mushroom is grown in shelves which may be of the same length or less and width range from 0.5m to 1.0m depending on the size of the fruiting bag

(4) *Prepration of medium for button mushroom*

- it is called composting.

- composting isa process whereby orgnic materials are connected into a stable medium for the growth

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Zinthu zomwe zimagwiliidwa ntchito popanga medium ya button mushroom .

The following materials are enough for production of 1 tonne of well dried cut maize stover or rice straw.

-20kg sulphate of ammonia

-20kg soya bean powder or cotton seed cake

-20kg white wash lime

-20kg molases

-30kg of single super phosphate

-30kg of rice bran

-150kg chicken maure

-10kg of calcitic lime

Phases of composting

Thwre are two phases

(1) outdoor composting*

-done outside the house under the shade

-involves wetting and mixing of compost ingredients

-done by cutting stover or straw into pieces of 3cm to 4cm in length

-30kg is enough to fill one square metre of bed

-it takes 16 days.

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(2)Indoor composting

-done inside production house

-aimed at pasteurising and conditioning the compost

-pasteurising is dne by introducing steam into the house

-it lasts for 12 to 14 days and is continuation of phase 1

Spawning

-is dne when moisture content of compost is 70%

-done at temperature of 25degrees and spawning rate is 400ml/m^2

-is done by spreading three quaters of required spawn quantity onto the compost and mixing it thoroughly

-doors n windows are are closed for the first 3days

-on day 4 ventilation is introduced omce in the morning and evening for 30minutes

- colonisation of the compost by mushroom fungus takes 12 to 14 days after spawning

Casing

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-the practice of adding moist soil onto the compost

-casing soil must be clay loam subsoil collected 10 to 13 cm deep

-Hp for this soil must be 7.0 to 7.5

-casing should be put in thicknes of 3 to 4cm and about 30litres of soil is required per square metre

-done 14 days after the compost is fully colonised by white mycelia

-casing soil shud always be moist inorder to maintain temperature of 16 + 22degrees

-at this condition pinheads will start forming from day 16 after casing.

THE IMPORTANCE OF CASING

-it promotes the formation of fruiting body and mycelia.

Pest and disease control of Button Mushroom.

PESTS

(1)Mushroom flies

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-They feed on mushroom fruiting body and Mycelia.

-it also transmit mites , eelworms and spores of other moulds

Control

-mantaining hygieneon the farm
.removing left over compost

-burn and burying sterms and pierces of mushroom.

(2)Mites

-Feeds on mushrom mycelia

Control

-Maintain cleanlines around the production house

(3)Rodents

-They feed on mycelia and mushroom fruiting body

-also makes holes in the casing and compost.

Control

-killing with traps or sticks

(4)Mollusk

Feed on mushroom direct

Control

-By hand picking physicaly

DISEASES OF BUTTON MUSHROOM

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(1)Dry Bubble

Symptoms

-pale brown spots on the cap of mushroom

Control

Practicing farm hygiene

-reducing air temperature and humidity in a production house

(2)Olive green Mould

Symptom

-Green grit like projections on the compost soon after pasteurisation

Control

-avoid too much moisture in the compost.

-proper pasteurisation of compost

-Avoid compaction and over pasteurisation of the compost.

(3)Mat

Symptoms

-Creamy to yellow mycelia fungual growth on both the compost and casing soil coating it completely.

Control

-maintaining farm hygene

-Avoid preparation of shallow casing

HARVESTING OF BUTTON MUSROOM

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Mushroom is ready for harvesting after 14+20 days after casing.

-done by holding the cup nd twist the mushroom then pull it togethr with the roots and cut off the roots and disporse them by burying.

Marketing of Button Mushroom

-mushroom ready for sale is clasified into different grades basing on their size and degree of maturity.

Grades of Button Mushroom ready for Sale..

(a)Buttons

-these may be either small or large size

-it is offered a highest price on the market

(b) Caps

-They are those whose their veils have opened or about to open.

-have medium value a the market

(c)Flats

-Are those with their veils opened and have flat caps and their gills are fully exposed .

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-flats have low demand on the market and hence fetch very low pricevat the market.